

Section 7.8: Communications Applications - Part 2

Back in section 7.1 we looked at some of the ways in which ICT helps us communicate with each other within our personal lives.

In this section we will build upon this and look at **work-related** communication methods including:

- | | | |
|----------------------------------|----------------------------|-----------------------------|
| 1 Internet | 2 Email | 3 Fax |
| 4 Electronic Conferencing | 5 Mobile Telephones | 6 Internet Telephony |

Key Concepts of this section:

- # Understand different ways that **ICT has affected how we communicate at work.**
- # Know various **uses of each example** of work-related communication methods.
- # Be able to **discuss each example** and **explain the uses** within a work-related situation.

Examples of Work Related ICT Communication Methods

The following examples explain how ICT-based communication methods effect how we work:

1 Internet

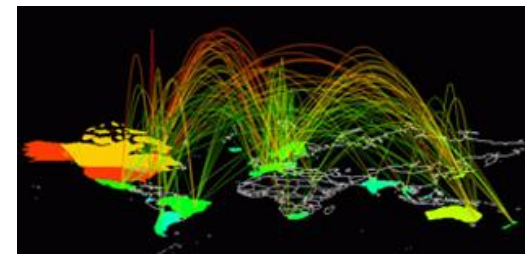
Overview of the Internet:

- # The Internet is a huge collection of '**Connected Networks**'. At time of writing (2nd Mar 2013) there are an estimated **5 billion computer devices connected to the Internet.**
- # Computing devices that can connect to the Internet include:
 - **Computers / Laptops**
 - **Mobile Phones**



Key Words:

Internet, Email, Fax, Electronic Conferencing, Mobile Telephones, Internet Telephony



The world is now connected via the Internet

- **Internet T.V's**
- **E-book readers (like Kindles)**
- **Digital Cameras**
- **etc.....**

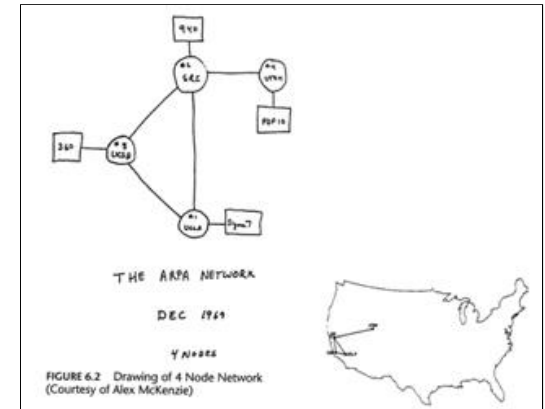
Each device needs '**Internet Connectivity**' capabilities, such as **wireless**, before it can be connected.

History of the Internet:

- # The first version of the Internet was **established in 1969**, where two computers were successfully connected between two Colleges in California.

By the end of 1971, there were 13 more sites connected to the Internet and it grew from there until it became the global network that we know today.

This early version was known as '**ARPANET**'.



Original design for **ARPANET**. Image shows a map for the area of California over which the first two computers would be connected.

Effects of the Internet (in terms of work-related communication):

Work-related uses:	
Websites -	Websites are hosted on the Internet and allow business to advertise globally . Customers access the sites using URL's. Some websites (like Amazon) allow users to purchase products directly online . This allows the company to operate without the huge costs of maintaining shops and stores .
Video Conferencing -	This allows large businesses with offices in many countries to hold meetings without having to fly out to visit each other. This is much cheaper and faster .
Email -	Email messages can be sent over the Internet almost instantly . This allows employees to communicate very easily.
File Transfer -	Files can be sent over the Internet very quickly. This means that documents, accounts etc can be transferred from office to office or to clients much quicker than manual methods (like postal delivery).



The **NEXT Computer**. Image shows the **first ever Web Server** (place to host websites)

NOTE:

The **Internet** is **NOT the same** thing as the **World Wide Web**.
WWW docs (e.g. websites) are hosted on special computers (called servers) which are connected to other computers via the Internet.

2 Email - (Electronic Mail)

Overview of Email:

Emails allow people to **exchange digital messages**. These messages can be **quickly sent to one person or to thousands** at the click of a button.

Files and images can also be **attached to an email** and transferred.

Emails can only be sent to people who have an **Email Address (e.g. louie1066@gmail.com)**. This information tells the email where it should go.

Emails are **sent across networks** (usually either a LAN or a WAN). For example:

- **WAN (Internet) - emails transferred across a large area via computers connected over the Internet.**
- **LAN (Local Area Network) - emails transferred over short distances, via a cable, to other computers connected to the same network in a building or a room.**

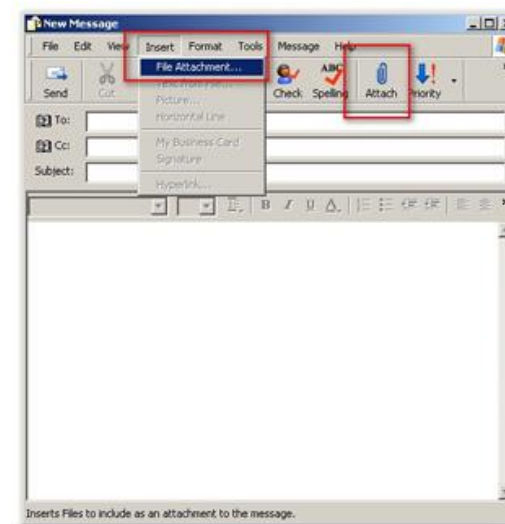


Ray Tomlinson

History of Email:

The first Internet email was sent in **1971** by a computer engineer named **Ray Tomlinson**. The contents of the first email was simply 'QWERTYUIOP'.

This first email was **transferred across ARPANET** (early version of the Internet).



Files can be attached to emails and sent to email addresses

Effects of Email (in terms of work-related communication):

Work-related uses:	
Fast Communication -	Emails are transferred instantly, even across large distances. This allows offices and employees to communicate even if they are in different areas or countries.
Easy Communication -	The same email can be sent to many people . For example - managers who want to send the same information to many employees, only need to type the message once .
File Transfer -	Files can be attached to emails and sent to email addresses. This means that business documents, contracts etc can be sent to employees or clients very fast.
Efficient Marketing -	Emails containing special offers or promotions can be sent to customers email addresses . This is much easier, quicker and

cheaper than having to market promotions via posters, t.v adverts and leaflets etc.

3 Fax

Overview of Fax:

Fax machines send **scanned images and text** across **telephone connections**.

The steps to sending a fax are as follows:

- **User A scans a printer document into the fax machine**
- **Fax machine converts the scanned document into a single image**
- **User A inputs the telephone number of the destination fax machine owned by User B**
- **Digital image is sent through the telephone network to the destination fax machine**
- **Destination fax machine receives the digital image and prints it onto paper**
- **User B reads the message fax 'd by User A.**



An old fax machine from the 1990's

Fax compared to Emails:

Fax		Email	
Paper can be wasted as you must scan and print physical documents.	🙄	No paper necessary as process can be entirely digital.	👍
Even small messages must be scanned and printed which takes time .	🙄	Email messages and files can be sent instantly if you have digital copies ready.	👍
Images sent between fax machines are much more secure than digital files sent over email.	👍	Files and messages sent over email can be intercepted and stolen.	🙄



Diagram showing how fax works

NOTE:

Fax machines have largely been replaced by more efficient methods of communication like emails.

4 Electronic Conferencing (Video Conferencing)

Overview of Video Conferencing:

- # Allows **two or more locations** to carry out **two-way video and audio connections**. This allows businesses to carry out meetings from any location in the world.
- # Two-way video and audio connections means that people from each location can see and speak to each other.

For example - Managers at an office in **New York** can hold a **face to face** meeting with managers at an office in **Tokyo** without having to travel.

- # To hold a video conference, everyone who wants to be part of the meeting will need the following hardware:

- **Broadband Internet - fast connection to handle the transmission of large video and sound data**
- **Computer**
- **Web cam - to capture and send video images**
- **Microphones - capture and send voices / sounds**
- **Monitors / Screens - to display the video images**
- **Speakers - to output the voices / sounds**



A video conference between 4 different locations



Video conferencing hardware include **router**, **web cam** and **screen**

Effects of Video Conferencing (in terms of work-related communication):

Work-related effects:	
Reduced Costs -	Video conferencing means that meetings can be held across long distances with no transport or accommodation costs. Potential recruits can also be interviewed face-to-face without having to pay travel expenses.
Time Efficiency -	If you have to travel to attend a meeting then this means you could be away from the office and your duties for several days. Video conference meetings can be held over far less time which



Video interviews can be carried out from any location with no travel expenses

	means that people can get back to their jobs quicker.
More Regular Meetings -	Meetings can be held more frequently than they could if traveling was required. This means that more business decisions and deals can be made.

5 Mobile Telephones

Overview of Mobiles:

- # These are **wireless devices** that allow users to **make telephone calls** from pretty much any location both indoors and outside.
- # Mobiles connect to the **telephone system** using **radio signals** and you must be within range of a mobile tower. These **mobile towers provide a boost to the radio signal** which increases the range over which calls can be made.
- # Modern mobiles have lots of useful features including:
 - **Send / receive voice telephone calls**
 - **Send / receive text messages**
 - **Send / receive pictures**
 - **Send / receive files**
 - **Email**
 - **Internet access**
 - **Digital camera**
 - **Business applications (Edit MS Office docs, Calendars, Organisers etc).**



Mobile Phone running Excel and a Mobile Tower which boost radio signals

History of Mobile Phones:

- # The first commercial mobile was released by **Motorola in 1983** and was named the **DynaTAC 8000X**.
- # The DynaTAC 8000X was very large and cost Almost \$4,000.

Effects of Mobiles (in terms of work-related communication):

Work-related uses:	
Improved Communication -	Employees, managers or owners are always available to speak to as long as they have their mobile phone with them. Customers can also contact business employees.
Organisation -	Mobile phones allow users to organise business contacts including phone numbers, email addresses and home addresses. They also allow users to plan their schedules using built in calendars . This helps users to meet deadlines etc.
File Transfer -	Business files can be transferred either from mobile to mobile using blue tooth or from mobile to email accounts using Internet connections .
Working on the go -	More powerful mobiles allow users to create or edit simple documents such as Word, Excel or even Access. This means that users can be productive even when out of the office.



The DynaTAC 8000X was the first commercial hand-held mobile phone

6 Internet Telephony (VoIP - voice over Internet protocol)

Overview of VoIP:

- # Internet Telephony services (**aka VoIP**) allow users to **talk to each other over the Internet**.

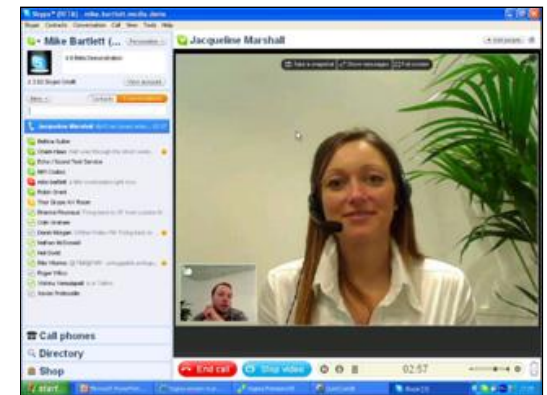
- # VoIP services are **usually free of charge** which means that users can even make International calls at no cost.
- # There are many Internet Telephony applications available. Some of them are **included with web browsers** and some are **stand-alone programs**. The list below shows you some of them:
 - **CoolTalk** - released in 1996 as a plug in for Netscape navigator 3.0
 - **NetMeeting** - Microsoft product bundled with early versions of Explorer
 - **MSN / Yahoo Messenger** - chat programs with VoIP capabilities
 - **Skype** - stand-alone VoIP program that is becoming very popular.



Windows NetMeeting

Effects of Internet Telephony (in terms of work-related communication):

Work-related effects:	
Reduced Costs -	Calls can be made to any location in the world for free of charge . This can save business lots of money when compared to the International costs of regular phone services and mobiles.
Video and File Transfers -	VoIP services can do extra stuff that the regular phone service cannot. For example you can hold video calls and transfer documents / files . These tasks cannot be handled by normal telephones.



More modern VoIP programs, such as Skype, are not bundled in browsers but are stand-alone products

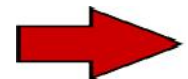
NOTE:

Many VoIP services also allow users to **transmit video images** and **transfer files**.



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